

3.3 Earthquakes present a hazard to many people and need to be managed carefully	a) Preparation for earthquakes (warning and evacuation, building design, remote sensing and GIS). (4) b) Short-term responses and relief (emergency aid, shelter and supplies). c) Longer-term planning (risk assessment, hazard mapping and rebuilding programmes). (5)
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Integrated skills	
(1)	Use world maps to show the distribution of different hazards.
(2)	Use a range of different maps to show links between tectonic boundary and hazard type.
(3)	Use social media sources, satellite images and socio-economic data to assess varying impacts.
(4)	Use GIS to investigate preparation for earthquake hazards.
(5)	Use online data sources to research the range of shorter and longer-term responses for a particular earthquake event.

1.4 Assessment of fieldwork skills – Section B

Fieldwork is assessed in Section B of Paper 1. Students are required to complete **one** geographical enquiry involving fieldwork relating to **one** topic in Paper 1.

Paper 1: Physical geography

- River environments.
- Coastal environments.
- Hazardous environments.

Centres must ensure that:

- Primary data collection includes quantitative and qualitative techniques.
- Secondary data collection includes the use of at least **two** different secondary data sources for your chosen environment.

Practical skills

As part of – and in addition to – undertaking the geographical enquiry, students should acquire and be able to apply the following skills:

- **graphical skills** – compiling graphs and flow lines, using proportional symbols, annotating maps, diagrams and photographs
- **map skills** (including use of digital maps) – using grid references, understanding scales, recognising symbols, identifying landforms and human features of the landscape
- **photo-interpretation skills** – reading vertical and oblique aerial photographs and satellite images, including GIS
- **sketching skills** – communicating ideas through simple sketch maps and field sketches
- **spatial awareness** – identifying the relative locations and relationships between features.

Cognitive enquiry skills

Students should acquire and be able to apply the following skills:

- **analysis of findings** – reviewing and interpreting quantitative and qualitative information using appropriate media
- **use of statistical skills** – simple descriptive statistics, such as lines of best fit, means, medians, modes, etc.
- **conflict resolution skills** – identifying the views of interested people (stakeholders), recognising that stakeholders may have strongly different attitudes and feelings towards a particular issue
- **evaluation of findings** – appraisal and review of data and information to see if they are accurate and suitable for the purpose, or misleading and unreliable.

Fieldwork questions will include questions set in a familiar and unfamiliar fieldwork context.

Questions set in a familiar fieldwork context will require students to interpret, analyse, evaluate and make judgements about their own fieldwork (AO3). They will also require students to communicate their findings (AO4).

Questions set in an unfamiliar fieldwork context will be set in in the fieldwork environment you have studied and will relate to the fieldwork investigation you have done (as set out in column 4 in the table below), they will however use unfamiliar fieldwork data and students will need to show that they can apply their fieldwork understanding and skills to interpret and analyse this data.

Contexts for fieldwork
Paper 1: Physical geography

Section B topic	Geographical enquiry	Suggested methods of primary and secondary data collection for familiar fieldwork contexts	What students need to learn for unfamiliar primary and secondary fieldwork contexts in Paper 1
River environments	Investigation of river processes and form through primary and secondary fieldwork evidence	Primary Quantitative e.g. (1) channel measurements - velocity, width, depth and gradient (2) measurements of sediment - size and shape Qualitative e.g. (1) annotated field sketches of the river channel and its features, (2) photographs to show how the channel changes downstream Secondary (1) A GIS topographic map, e.g. from ArcGIS Online or Google Earth, (2) local secondary data on river flows or regimes	Primary Quantitative • River channel characteristics: width, depth and velocity • River gradient Qualitative • Annotated field sketches Secondary • GIS topographic map
Coastal environments	Investigation of coastal processes and form through primary and secondary fieldwork evidence	Primary Quantitative e.g. (1) sediment size and shape measurements, beach profile survey, (2) measurement of erosional features - a cliff or intertidal zone Qualitative e.g. (1) annotated field sketches of particular coastal features, (2) photographs to show how there are variations along a stretch of the coast.	Primary Quantitative • Sediment: size and shape • Beach profile Qualitative • Annotated field sketches Secondary • Local geology map

Section B topic	Geographical enquiry	Suggested methods of primary and secondary data collection for familiar fieldwork contexts	What students need to learn for unfamiliar primary and secondary fieldwork contexts in Paper 1
		Secondary (1) A local geology map (paper or digital), (2) local secondary data on coastal change, e.g. historic maps.	
Hazardous environments	Investigation of physical processes involved in an extreme weather event through the recording of primary and secondary fieldwork evidence	Primary Quantitative e.g. (1) recording of a weather diary and local risk and hazard maps, (2) structured questionnaire about hazard perception Qualitative e.g. (1) annotated field sketches to show evidence of an extreme weather event, (2) photographs / video taken before, during and after the extreme weather event Secondary (1) An online feed from a local weather station, e.g. Wunderground, (2) local secondary data on weather events, e.g. newspapers, online accounts.	Primary Quantitative - Weather diary: wind, rain, temperature, and air pressure - Local risk and hazard mapping Qualitative - Annotated field sketches Secondary: - Local live feed of an extreme weather event.